

Slam EX5

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[Github](#)

5.1:

- a) A standard extrinsic camera matrix $[R|t]$ transforms a 3D point from the global (world) coordinate frame to the camera coordinate frame:

$$X_{\text{camera}} = R * X_{\text{global}} + t$$

GTSAM's `gtsam.StereoCamera/gtsam.Pose3` requires the inverse transformation,

which maps a point from the camera's local coordinate system back to the global (world) coordinate system. We isolate X_{global} mathematically:

$$X_{\text{camera}} - t = R * X_{\text{global}}$$

$$R^{-1} * (X_{\text{camera}} - t) = X_{\text{global}}$$

$$X_{\text{global}} = R^T * X_{\text{camera}} - R^T * t \quad (\text{Since } R \text{ is orthogonal, } R^{-1} = R^T)$$

Therefore, the transformation parameters from the camera coordinate system to the global coordinate system are:

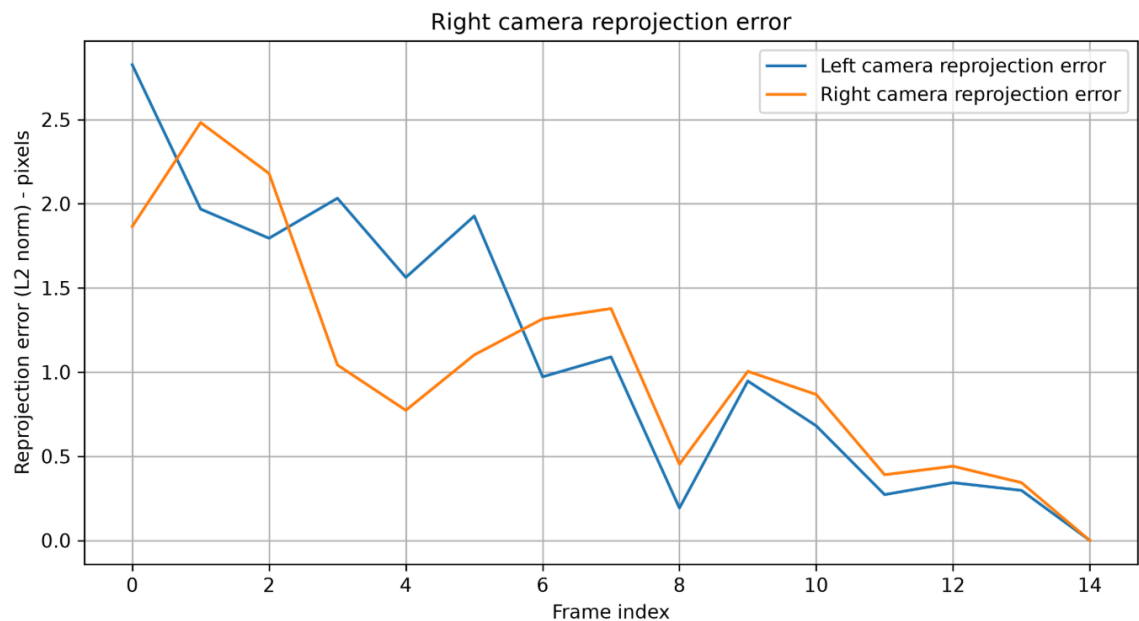
- Rotation (R_{c2w}): R^T

- Translation (t_{c2w}): $-R^T * t$ (This represents the camera center 'C')

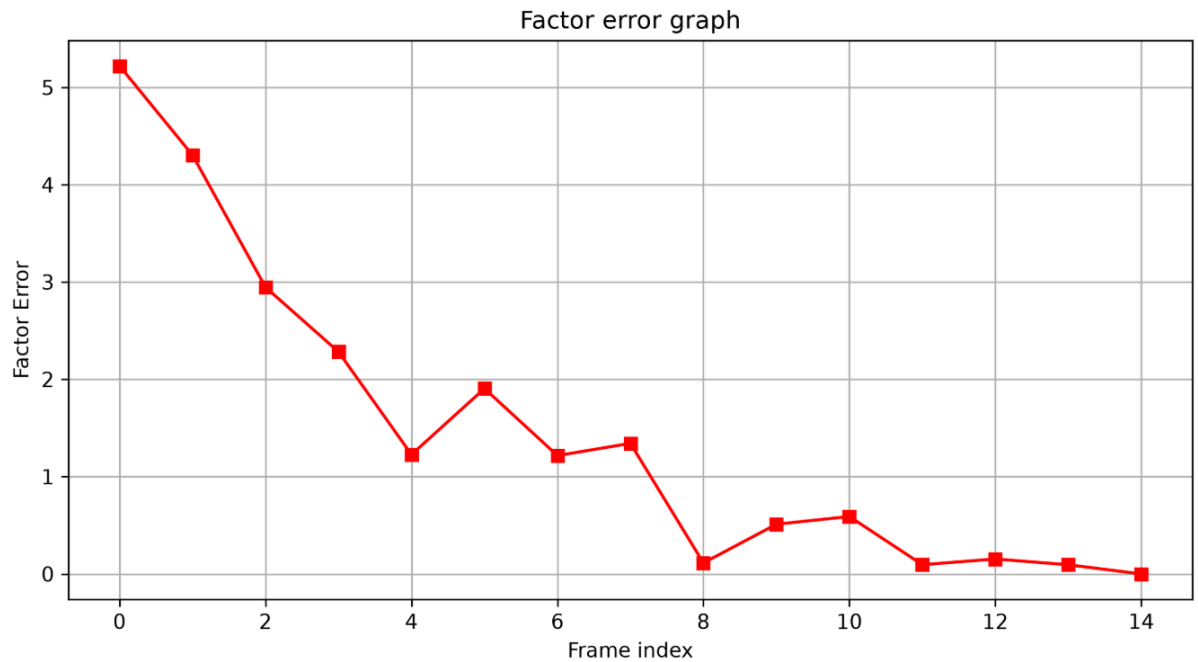
Hence :

$$[R^T | -R^T t]$$

b)



c)



d) We utilized an Isotropic noise model with a pixel standard deviation (sigma) of 1.0 pixel across all three stereo measurement channels (u_{left} , u_{right} , v):

$$\text{Sigma} = \begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 1.0 \end{bmatrix} = I_{(3 \times 3)}$$

Reasoning:

Since feature tracks are extracted using calibrated interest-point detectors (e.g., ORB) and filtered for rectified epipolar alignment, the measurement errors are modeled as independent and identically distributed (i.i.d.) Gaussian white noise with an uncertainty bound of 1.0 pixel.

e) In GTSAM, a non-linear factor's numerical error is defined as half of the squared Mahalanobis distance of the measurement residual vector 'e':

$$\text{Error_factor} = 0.5 * e^T * \text{Sigma}^{-1} * e$$

Where 'e' is the stereo reprojection residual vector:

$$e = \begin{bmatrix} u_{L_proj} - u_{L_meas} \\ u_{R_proj} - u_{R_meas} \\ v_{proj} - v_{meas} \end{bmatrix}$$

The geometric Reprojection Error (L2 norm) is defined as:

$$\|e\|_2 = \sqrt{(u_{L,proj} - u_{L,meas})^2 + (u_{R,proj} - u_{R,meas})^2 + (v_{proj} - v_{meas})^2}$$

Since our chosen covariance matrix is isotropic with $\sigma = 1.0$, its inverse

simplifies directly to the Identity matrix ($\text{Sigma}^{-1} = I$). Substituting this:

$$\text{Error}_{factor} = \frac{1}{2} e^T I e = \frac{1}{2} \|e\|_2^2$$

Hence, the factor error as a function of the geometric reprojection error is:

$$\text{Error}_{factor} = \frac{1}{2} \cdot (\text{Reprojection_Error})^2$$

5.2:

The projection equations for a world point into cameras A and B are given by:

$$1. X_A = R_A X_{world} + t_A$$

$$2. X_B = R_B X_{world} + t_B$$

To find the direct relative transformation $T_{(A \rightarrow B)}$, we substitute the isolated expression for X_{world} from equation (1) into equation (2):

$$X_B = R_B [R_A^T (X_A - t_A)] + t_B$$

$$X_B = (R_B R_A^T) X_A + (t_B - R_B R_A^T t_A)$$

Thus, the relative rotation and translation defining $T_{(A \rightarrow B)}$ are:

$$R_{A \rightarrow B} = R_B R_A^T$$

$$t_{A \rightarrow B} = t_B - R_B R_A^T t_A$$

The extrinsic camera matrix of camera B inside the local reference frame of camera A is exactly this relative projection layout:

$$[\mathbf{R}_{A \rightarrow B} \mid t_{A \rightarrow B}] = [\bar{R}_B \bar{R}_A^T \mid t_B - R_B \bar{R}_A^T t_A]$$

5.3:

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SECTION 5.3: FIRST BUNDLE ADJUSTMENT WINDOW (10 FRAMES)
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Bundle Window spans from Frame 0 to Frame 9 (Total 10 frames)
Number of landmarks participating in this bundle: 830

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FACTOR GRAPH STATISTICS
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Total Factors in Graph (including prior):      2365
- Prior factors (anchoring start pose):        1
- Stereo projection factors:                   2364
- Landmarks inserted into graph:              309

Total Factor Graph Error   BEFORE Optimization: 2749.0429
Average Factor Error      BEFORE Optimization:  1.1624

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LARGEST INITIAL ERROR FACTOR – INITIAL VALUES
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Factor Index in Graph:      422
Camera Frame Key (c):       symbol('c', 9) [GTSAM key = 7133701809754865673]
Landmark Point Key (q):     symbol('q', 97) [GTSAM key = 8142508126285856865]
Initial Factor Error:       16.051786

Measurement  – Left X:   315.60 | Right X:   290.40 | Y:   142.80
Projection   – Left X:   320.87 | Right X:   297.73 | Y:   142.77

L2 Pixel Distance from Measurement (Left camera): 5.2728 px
L2 Pixel Distance from Measurement (Right camera): 7.3256 px

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RUNNING LEVENBERG-MARQUARDT OPTIMIZATION...
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Optimization Complete!

Total Factor Graph Error   AFTER Optimization:  613.1754
Average Factor Error      AFTER Optimization:   0.2593

Error Reduction:  2749.0429 -->  613.1754 (reduced by 77.7%)

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LARGEST INITIAL ERROR FACTOR – OPTIMIZED VALUES
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Same Factor:  symbol('c', 9) x symbol('q', 97)
Optimized Factor Error:      0.252585 (was 16.051786)

Measurement  – Left X:   315.60 | Right X:   290.40 | Y:   142.80
Projection   – Left X:   315.70 | Right X:   291.08 | Y:   142.99

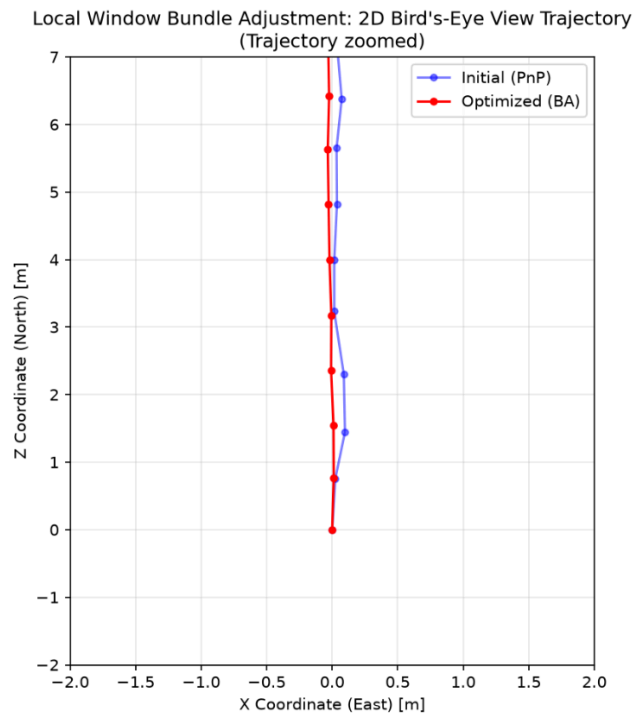
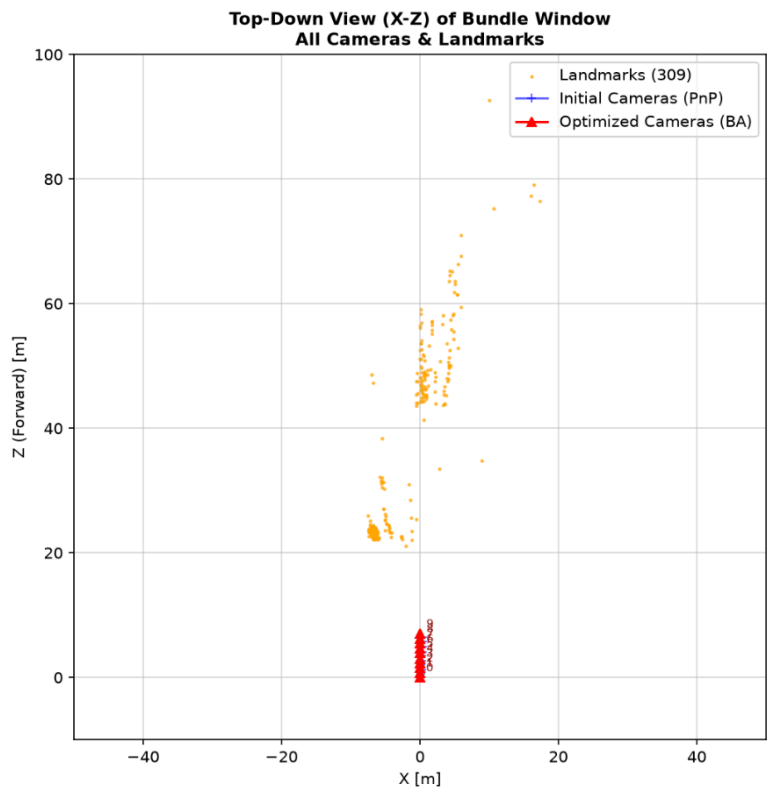
L2 Pixel Distance from Measurement (Left camera): 0.2126 px (was 5.2728 px)
L2 Pixel Distance from Measurement (Right camera): 0.7041 px (was 7.3256 px)
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Left:

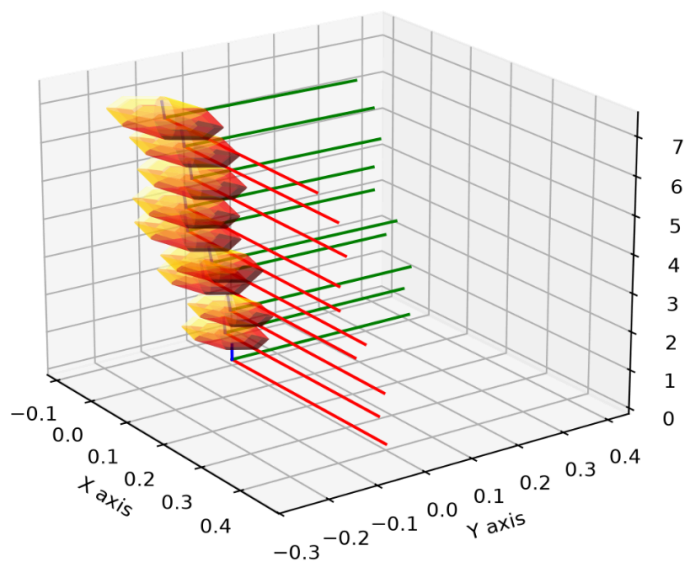


Right:

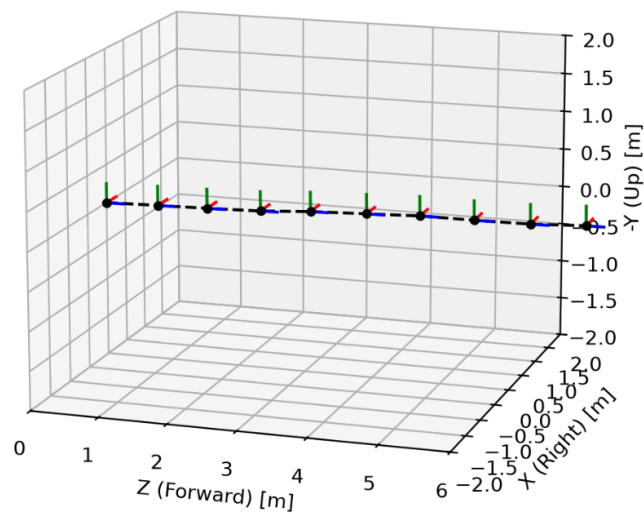




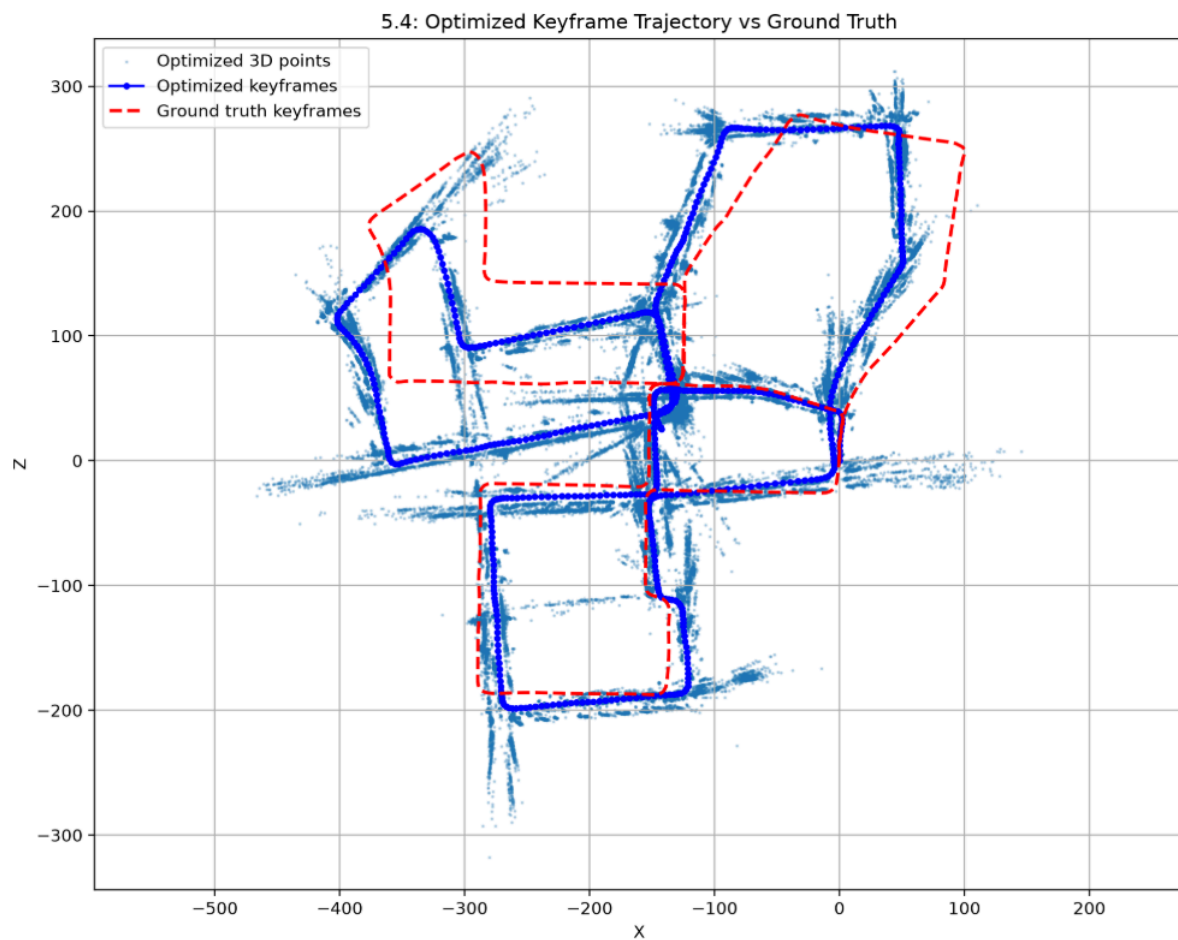
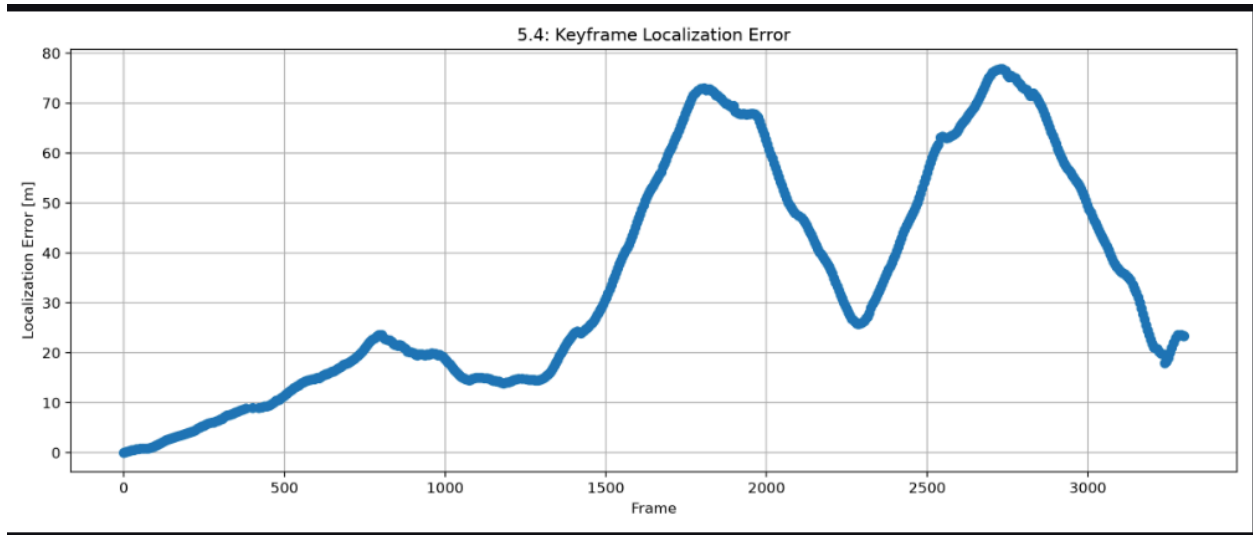
Plot Trajectory
GTSAM Factor Graph State with Marginal Covariances



Local Window Bundle Adjustment: 3D Optimized Trajectory



5.4:



The optimized keyframe trajectory successfully captures the overall structure of the KITTI sequence. The estimated trajectory follows the main geometric layout of the route, including its major turns, straight segments, and loop-like structures. While the local bundle adjustment produces accurate pose estimates within each optimization window, the global trajectory is reconstructed by chaining relative poses between consecutive keyframes. As a result, small local estimation errors accumulate over long distances, leading to noticeable drift in the global trajectory.

This behavior is reflected in the localization error curve. The error remains relatively low during the early stages of the sequence, gradually increases as drift accumulates, reaches a maximum of approximately **136 meters**, and later decreases as the trajectory geometry becomes more consistent with the ground truth. Over the entire sequence, the mean localization error is approximately **52 meters**.

Despite this accumulated drift, the estimated trajectory preserves the correct global scale (estimated-to-ground-truth step ratio ≈ 0.99) and remains geometrically consistent with the overall shape of the ground-truth route. These results demonstrate that local sliding-window bundle adjustment significantly improves local pose estimation accuracy, while also highlighting the limitations of purely local